

Private AI Code Assistant: Enterprise GPU Sizing Report v2

April 2026 — Full-stack TCO including Red Hat OpenShift subscriptions, OpenShift AI, and infrastructure nodes

Executive Summary

Infrastructure sizing and total cost of ownership for deploying a fully private, self-hosted AI coding assistant on Red Hat OpenShift — targeting regulated industries (banking, government, healthcare) requiring complete data sovereignty and IP control.

\$18-41

Per Dev/Month (Full Stack)

6

Models Evaluated

20 tok/s

Throughput Baseline

~20%

Peak Concurrency

Architecture



Key Findings

Best Value — Full-Stack \$18/dev/mo at 200 Developers

Qwen3.6-35B-A3B on L40S (g6e.xlarge) with full Red Hat stack including OpenShift subscriptions, OpenShift AI, Hosted Control Plane, and 3 infra nodes. At 100 devs: \$29/dev/mo. At 50 devs: \$41/dev/mo. These are complete costs — no hidden per-seat licensing.

Architecture Advantage — MoE + DeltaNet

Hybrid MoE models reduce KV cache by 75-95% vs standard transformers, enabling 4-10x more

concurrent users per GPU. VRAM is always the binding constraint for MoE — bandwidth is never the bottleneck.

Memory Efficiency — PagedAttention + Prefix Caching

vLLM PagedAttention allocates KV dynamically and frees it instantly on completion. llm-d routes multi-turn conversations for prefix reuse. Together: 2.5-4x effective concurrency beyond worst-case limits.

Compatibility Trade-off

DeltaNet hybrid models (highest quality) require NVIDIA CUDA. Alternative accelerators (Trainium, TPU) are limited to standard MoE architectures with lower code quality scores.

The Case for Private AI Code Assistance

Regulated industries face an urgent need for AI developer productivity tools, but regulatory and security requirements prevent adoption of SaaS-based solutions.

Market Drivers

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Driver	Deadline / Status	Impact on AI Coding
Private AI Adoption	\$109B global spend (2024)	55% to 78% enterprise adoption

Customer Segments

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Segment	Regulatory Framework	Key AI Coding Concern
Telecom / CI	NIS2 Directive	Supply-chain security; controlled CI/CD and model chain

SaaS vs Private Deployment

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Dimension	SaaS AI Coding	Private Red Hat Stack
Cost scaling	Linear per seat (\$19-39/mo)	Sub-linear — tracks concurrency, not headcount

Fines Up To EUR 35M / 7% Global Turnover

EU AI Act non-compliance can compound with GDPR and DORA penalties. Architecturally isolated deployment eliminates the regulatory risk of third-party AI data processing.

Solution Advantages: Red Hat AI Platform

Cost Advantages

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Capability	Technical Detail	Cost Impact
FP8 quantization	2x parameter density with <1% quality loss	Smaller GPUs viable for same model

Full-Stack Cost: Private vs SaaS

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Team	Full-Stack Private (3yr SP)	SaaS Enterprise (\$39/dev)	Annual Savings
200 devs	\$3,662/mo (\$18/dev)	\$7,800/mo	\$50K/yr savings

Private stack includes EC2 GPU + infra + OpenShift subscription + OpenShift AI + Hosted Control Plane. SaaS at \$39/dev/mo list (Copilot Enterprise).

Security & Data Sovereignty

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Risk	SaaS AI	Private Red Hat
Audit trail ownership	Vendor-held; export limited	Full ownership; customer SIEM

Operational Advantages

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Task	DIY K8s + vLLM	Red Hat AI Platform
Support	Community only	24x7 Red Hat enterprise support

Model Specifications & Benchmarks

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Model	Architecture	Params (Total/Active)	KV/Token	FP8 Size	vLLM
Nemotron Nano 30B	Mamba-2 + MoE	~30B / 3.3B	~4 KB	~30 GB	Yes

Green: recommended. Blue: highest quality (dense).

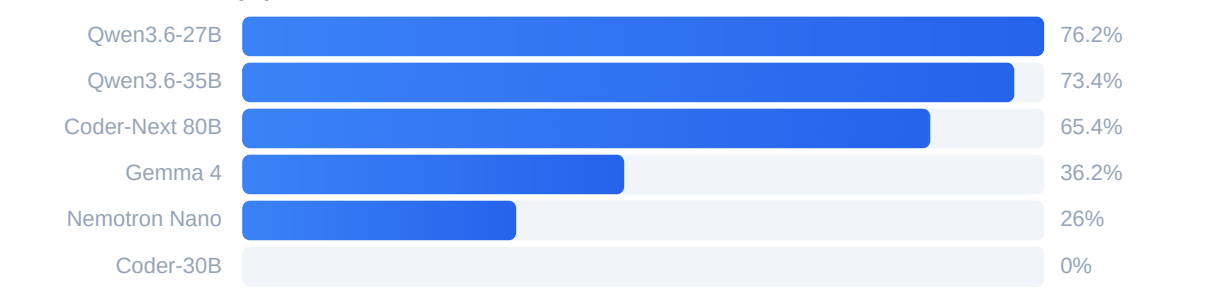
Code Quality Benchmarks

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Model	SWE-bench Verified	SWE-bench Pro	LiveCodeBench v6	HumanEval
Nemotron Nano 30B	26.0%	--	38.5%	82.3%

SWE-bench Verified Scores

SWE-bench Verified (%)



Code Quality Ranking

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Rank	Model	SWE-bench V.	Assessment	Deployment Trade-off
#6	Qwen3-Coder-30B-A3B	--	Weakest coverage	Only model for Trainium/TPU

Optimal Choice

Qwen3.6-35B-A3B delivers 95% of top-ranked dense model quality at 9x lower compute — the optimal balance of quality, concurrency, and cost.

GPU Specifications (us-east-1)

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GPU	VRAM	Mem BW	FP8	Instance	vCPUs	OD \$/hr	3yr SP \$/hr
B200	192	8,000	9,000	p6-b200.48xlarge	192	\$113.93	\$91.14
192GB	GB	GB/s	TF	(8x)			

Instance Size Optimization

Single-GPU instances minimize both GPU waste and OpenShift subscription costs (fewer vCPUs to license).

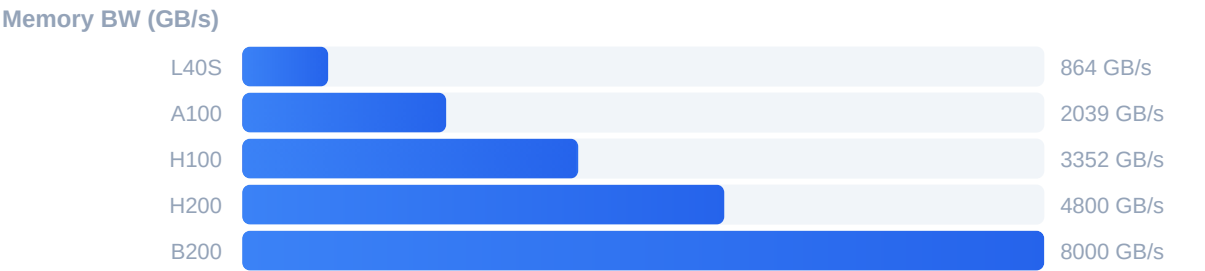
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Instance	GPUs	vCPUs	Idle if 1 GPU Needed	Wasted \$/mo
p5.48xlarge	8	192	7 GPUs	\$17,570

Key Insight

L40S and H100 offer single-GPU instances. A100, H200, and B200 force 8-GPU purchases with 192 vCPUs — dramatically increasing OpenShift subscription costs even if only 1 GPU is needed.

Memory Bandwidth



VRAM Capacity Analysis

Maximum concurrent sessions per single GPU. FP8 weights + FP8 KV cache.

Qwen3.6-35B-A3B (Recommended)

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Context	KV/Req (GB)	L40S	A100	H100	H200	B200
256K	2.52	4	17	17	41	62

Qwen3-Coder-30B-A3B

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Context	KV/Req (GB)	L40S	A100	H100	H200	B200
256K	12.00	1	3	3	9	13

Coder-Next 80B (H200+ only)

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Context	KV/Req (GB)	L40S	A100	H100	H200	B200
256K	3.00	--	--	--	19	36

Gemma 4 27B-A4B

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Context	KV/Req (GB)	L40S	A100	H100	H200	B200
256K	27.50	--	1	1	4	5

Max Concurrent Users at 64K (by GPU)

Qwen3.6-35B

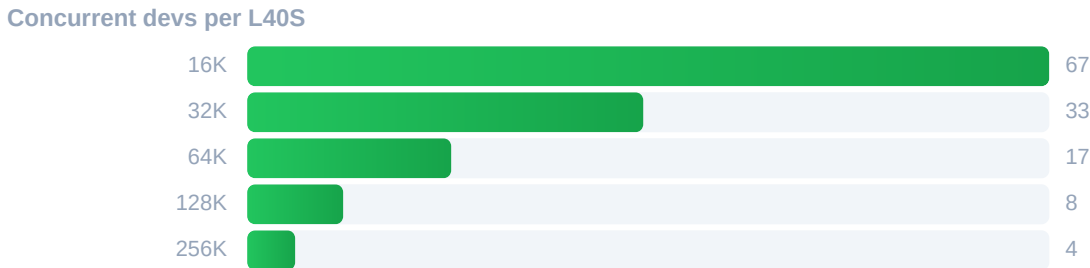


Coder-30B





Context Window Impact (Qwen3.6-35B on L40S)



Doubling context halves concurrency. 64K accommodates most agentic workflows.

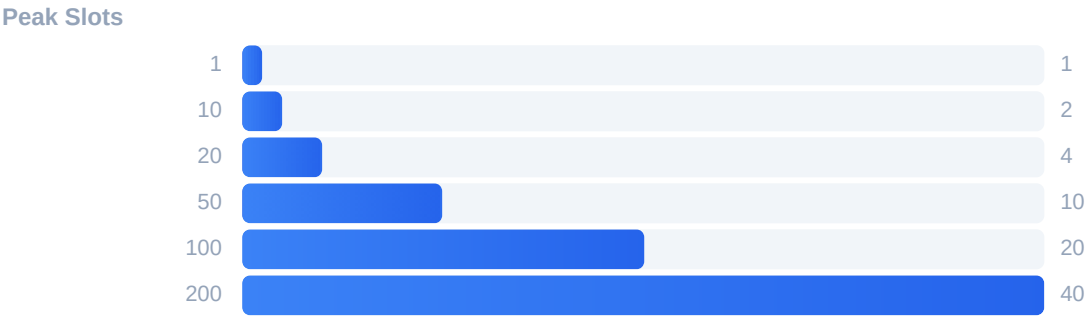
Concurrency & Throughput

Workday Concurrency Model

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Factor	Value	Derivation
Design factor	~20%	$0.163 \times 1.20 = 0.20$

Peak Concurrent Slots by Team



Why 20 Tokens/Second Baseline

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Throughput	Developer Experience	Impact
> 40 tok/s	Near-instantaneous	None

Reading Speed Context

Average code reading speed is ~3-5 tokens/second. At 20 tok/s, output is 4-6x faster than reading. For MoE models, 20 vs 30 tok/s produces identical GPU requirements (VRAM-bound, not bandwidth-bound).

PagedAttention & Prefix Caching

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Technique	How It Works	Concurrency Impact
Cross-user sharing	Identical system prompts share KV pages	N users x 1 copy, not N copies

Conservative Sizing

GPU counts in this report represent worst-case peak load. Real deployments typically need 20-40% fewer GPUs.

Recommendations — Three Perspectives

All recommendations use Qwen3.6-35B-A3B at 64K context with FP8 quantization, 20 tok/s sustained baseline.

Perspective A: Best Value (Lowest \$/dev/mo)

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Team	Peak Slots	GPU Config	Instance	GPU \$/mo (3yr SP)
200 devs	40	3x L40S	3x g6e.xlarge	\$2,037

17 concurrent users per L40S at 64K. GPU cost only — see Full-Stack TCO for complete pricing.

Perspective B: Best Code Quality

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Team	Model	Instance	Slots	GPU \$/mo (3yr SP)
Any ≤200	Coder-Next 80B	1x p5en.48xlarge (8x H200)	~776	\$35,442

Premium Tier

p5en.48xlarge has massive excess capacity for ≤200 devs. No smaller H200 instance exists. 192 vCPUs drive high OpenShift subscription costs (\$3,002/mo OCP + \$3,470/mo OCP AI). Total full-stack: \$42,461/mo.

Perspective C: Best Balance (Recommended)

Qwen3.6-35B-A3B on L40S

73.4% SWE-bench — only 2.8pp below the dense 27B — at a fraction of the premium option cost. Same infrastructure as Perspective A. This is the recommended path for most organizations.

Context Length Impact (L40S GPUs Needed)

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Team	Peak Slots	32K	64K	128K
200 devs	40	2x L40S	3x L40S	5x L40S

Most agentic workloads operate within 32K-64K effective context.

Full-Stack Total Cost of Ownership (3-Year)

Complete deployment cost including GPU compute, infrastructure, OpenShift subscriptions, OpenShift AI, and Hosted Control Plane. 3-year No Upfront Savings Plan pricing.

Cost Components (Per Cluster)

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Component	Description	Pricing
OpenShift AI	AI platform license, all worker nodes	\$0.022/vCPU/hr

Region: US East (N. Virginia) / us-east-1. Same subscription model for ROSA (AWS) and OSD (GCP).

Best Value: Full-Stack TCO (Qwen3.6-35B on L40S, 64K)

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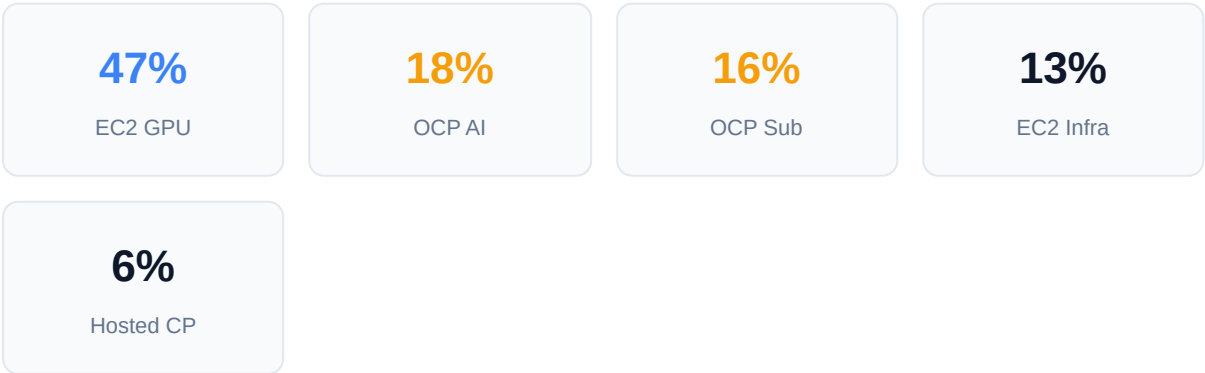
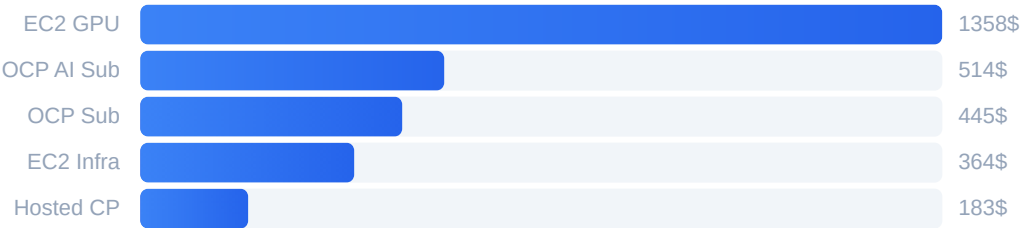
Component	1 dev	10 devs	20 devs	50 devs	100 devs	200 devs
3-Year TCO	\$74,340	\$74,340	\$74,340	\$74,340	\$103,104	\$131,832

Economies of Scale

Platform overhead (HCP, infra, subscriptions) is ~\$1,386/mo fixed. Adding 150 more developers (50 to 200) only increases cost by \$1,597/mo (from \$2,065 to \$3,662). Per-developer cost drops from \$41 to \$18.

Cost Component Breakdown (100 Devs, 64K)

Monthly Cost (\$)



GPU compute is 47% of total. Platform subscriptions (OCP + OCP AI) are 34% combined.

Context Length Impact on TCO

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Team	32K Total / Dev	64K Total / Dev	128K Total / Dev
200 devs	\$2,864 / \$14	\$3,662 / \$18	\$5,259 / \$26

Premium Quality TCO (Coder-Next 80B on H200)

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Component	1-100 devs	200 devs
Per Dev @ 200	--	\$212

10-15x Cost Premium

Premium quality (Coder-Next 80B) costs \$425/dev vs \$29/dev at 100 developers. The 192-vCPU p5en.48xlarge drives \$6,655/mo in platform subscriptions alone. Justified only when code quality definitively outweighs cost.

GPU-Only vs Full-Stack TCO (3-Year)

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Team (64K)	GPU-Only 3yr	Full-Stack 3yr	Platform Overhead %
200 devs	\$73,332	\$131,832	80%

Platform overhead is ~\$50K over 3 years (fixed). Percentage drops as GPU spend scales.

Phase 2: Alternative Accelerators

Model Limitation

DeltaNet hybrid models (highest quality) require CUDA. Alternatives limited to Qwen3-Coder-30B-A3B (standard MoE) — lower benchmarks than Qwen3.6-35B-A3B.

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Accelerator	Memory	Instance	Slots @ 64K	\$/hr	Platform
TPU v7	192 GB	4-chip VM	212	\$24.00	OSD (GCP)

Cost Comparison: GPU-Only (100-Dev Team, 64K)

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Option	Model	Instance	Monthly
Trainium2	Coder-30B (Std MoE)	1x trn2.48xlarge	\$26,105

L40S reference uses 3yr SP with superior DeltaNet model. Alternatives at on-demand pricing with standard MoE. L40S wins on both cost and quality.

Appendix A: 30 tok/s Baseline

Key Finding

For Qwen3.6-35B-A3B (MoE), 20 vs 30 tok/s produces identical GPU requirements. VRAM is always the bottleneck, not bandwidth.

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GPU	BW Limit @ 30 tok/s	VRAM Limit	Effective	Binding
B200	419	248	248	VRAM

Bandwidth limit far exceeds VRAM limit in all cases. TCO tables are valid for both 20 and 30 tok/s.

AWS pricing: us-east-1 (April 2026). OpenShift: 3yr plan at \$667/4-vCPU/yr. OCP AI: \$0.022/vCPU/hr. HCP: \$0.25/hr flat. 3x m5.2xlarge infra nodes per cluster. Validate with Red Hat sales and empirical benchmarks before procurement.